How Price-Aware Are AI Pricing Agents? Rival Price Cuts, Legal Price Rules and False Punishment Signals

Martin Kaiser, individual contributor. Research conducted at the Apart Research AI Collusion Resear

Abstract

Collusion audits of AI pricing agents look for punishment: after a rival's price cut, the agent stays low, then recovers. Real prices often pass through legal rules that hold or delay price changes, such as platform offer holds and Germany's 2026 rule that fuel prices may rise only at noon. We ask whether such rules produce the punishment pattern alone, and whether logging proposed next to published prices separates the model from the rule. In a synthetic duopoly with a scripted rival, we fork a language-model pricing agent (Qwen3-235B), impose a one-period rival price cut, increase or no change, and compare immediate execution with a three-period hold and a noon-style rule. The hold adds 0.159 of the Nash-to-monopoly gap (95% CI 0.107–0.212) to how long the agent's published price stays low after the cut. Its own proposals also stay lower (+0.125), consistent with anchoring on its held-down price. Replaying its unconstrained proposals through the rule explains only 0.066; the agent's adaptation adds 0.093 (0.056–0.134). The noon rule shows the same pattern at half the size. Proposal logs remove the rule's mechanical effect but cannot show how the model would behave without it. Audits should record rule state and test agents under the rules they face. Across 16 open models from Chinese and US developers, responses to the cut range from none to over twice the best response, and the hold produces a false punishment signal in 6 of 10 models tested. Results concern audit validity in a synthetic market.

1. Introduction

AI agents make more and more pricing decisions, and they are often not observed. We need to understand better how they behave, and we need auditing methods that work.

Software already sets many prices, and large language models make capable pricing agents cheap enough for any small business. Competition authorities worry that such agents could keep prices high without any agreement (Calvano et al. 2020; Fish et al. 2024; CMA 2026). Explicit price fixing is illegal, and independent parallel pricing generally is not. The practical question for an auditor is therefore one of evidence: does an agent sustain high prices by punishing rivals who undercut it, or is it simply responding to the market?

The standard test is a deviation audit. The auditor makes a rival cut its price once and watches the agent. A competitive agent responds briefly and follows the rival back. An agent that enforces high prices stays low for several periods, sacrificing profit to discipline the rival, and then recovers (Abreu 1986; Calvano et al. 2020). Recent work runs this test on LLM agents by forking a market at a checkpoint (Arslan et al. 2026; Tran and Yu 2026).

These audits treat the agent's decision and the market price as the same thing. In deployed systems they often are not. Booking platforms hold an approved offer for a fixed period. Western Australia requires fuel stations to fix each day's price 24 hours ahead. Since 1 April 2026, German fuel stations may raise prices only once a day, at noon, while cuts are allowed at any time; Austria has had a similar rule since 2009 (Bundestag 2026). Each rule makes a price cut last longer than the agent intended. To an auditor who sees only published prices, that looks like punishment.

We study this attribution problem directly. Our contributions:

1. **An audit design with an explicit execution layer.** We log proposed prices, published prices and rule state, and add an open-loop replay that separates the rule's mechanical effect from the agent's reaction to it. Scripted controls with known answers validate the harness (27 tests).
2. **Evidence that legal price rules change an LLM agent's behaviour, not just its published prices.** With a three-period hold, Qwen3-235B's own proposals stay lower after a rival cut ($\theta = 0.125$); the mechanical hold explains 41% of the published-price effect. A noon rule modelled on Germany's produces the same pattern at about half the size.
3. **A panel of 16 open models.** In the same market with the same prompt, the reaction to a rival's cut ranges from none to more than twice the best response, and the hold produces a false punishment signal in 6 of 10 models tested. The model family matters more than the developer's country.
4. **Audit and policy recommendations**: log proposals, published prices and rule state together, test each agent under the rules it faces, and treat a punishment pattern as evidence only after the execution layer is accounted for.

2. Related Work

Collusive strategies and their signatures. Collusion is sustained by the threat of punishment: grim trigger, short stick-and-carrot punishments (Abreu 1986, 1988), or price wars after low observed demand (Green and Porter 1984). Q-learning pricing agents learn reward-and-punishment schemes with recovery (Calvano et al. 2020; Klein 2021), and LLM agents reach supra-competitive prices that depend on prompt wording (Fish et al. 2024). The signature is ambiguous: menu costs, sticky adjustment (Noel 2007) and Edgeworth cycles (Maskin and Tirole 1988) also produce persistence and recovery without any strategic intent. A punishment test therefore needs a competitive benchmark (Harrington 2008).

Deviation audits of LLM agents. Arslan et al. (2026, App. E) apply forced deviations to prompted LLM pricing agents and show the verdict depends on the best-response benchmark. Tran and Yu (2026) clone markets counterfactually with myopic placebos and positive controls. Moraes and De-Losso (2026) argue that deviation–response–recovery paths alone do not establish punishment. In all three, the proposed and executed price coincide. We add the execution layer.

Price-timing rules. Restricting when prices may change is an active policy tool with mixed evidence. Under Perth's 24-hour timing law, pricing matches Maskin–Tirole commitment equilibria (Wang 2009), and firms learned to coordinate through price leadership and focal points (Byrne and de Roos 2019). Austrian-style rules can shift prices over time without lowering them (Obradovits 2014); empirical and laboratory evidence is mixed (Dewenter and Heimeshoff 2012; Haucap and Müller 2012). Early evidence on Germany's noon rule shows petrol margins rose by about 5–6 cents per litre, possibly by making prices more predictable (Jung et al. 2026). None of this work

considers AI pricing agents or audit validity.

Execution feedback and anchoring. In reinforcement learning, an execution layer that repeats the previous action changes what the agent learns from ("sticky actions", Machado et al. 2018). LLMs show anchoring on salient numbers that chain-of-thought does not remove (Lou 2024; Huang et al. 2025). Our agent sees only executed prices, so a held price may act as an anchor.

3. Methods

Market. Two firms sell differentiated products to a fresh group of customers each period, with demand q_i = max(0, 35.4 − p_i + 0.6 p_j) and marginal cost 15. The static Nash price is 36 and the joint-profit price 51.75; we normalise price effects by the gap G = 15.75. Prices lie on a €0.50 grid from 15 to 70. There is no inventory, so a cut does not change future opportunity costs.

Execution rules. The agent proposes a price; an executor decides what is published. - *Immediate* (no rule): every proposal is published. - *Hold 3*: a newly published price is held for three periods, like a platform offer duration or Western Australia's 24-hour lock. Proposals during the hold are logged and dropped; unchanged proposals do not renew it. - *Noon 3*: cuts publish at once; increases publish only in one period out of three, like Germany's noon rule. The noon slot is varied across histories.

Agents. The main LLM agent is Qwen3-235B-A22B-Instruct-2507 (Nebius Token Factory, temperature 0.7); the model panel in Section 4 uses the same prompt and settings for other open-weight models, with a larger token budget for reasoning models. Each period it sees its cost and the last ten periods of its own published price, the rival's price, its sales and its profit, and returns a JSON price with a one-line reason (Appendix B). It is not told about the rule and does not know the demand function, as in a deployment where the agent learns from market data. Scripted controls with known behaviour: *fixed* price (42); *myopic* exact best response to the last rival price; *sticky* partial adjustment towards the best response, anchored on its own last proposal (λ □ {0.3, 0.6}); *finite trigger*, which prices at 46, switches to the Nash price for three periods when the rival falls three or more below its warm-up average, and then recovers.

Fork audit. We construct 16 market histories with seeded rival base prices b ~ U[38, 48]. After eight warm-up periods under immediate execution, the agent chooses its checkpoint price. We then fork just before feedback. The rival's executed price is replaced by b − 8 (cut), b + 8 (raise) or b (no-op), and both firms' sales and profits are recomputed. Afterwards the rival returns to b in every branch. Each branch runs eight more decisions under each rule, switched on at the checkpoint with no active hold. LLM branches are repeated four times. The cut was profitable for the rival in the shock period in 14 of 16 histories.

Estimands. For rule d, R_d is the mean over decisions h = 2..5 after the shock of (p_no-op − p_cut)/G; h = 1 is the first decision that sees the shock. The rule effect is θ_d = R_d − R_none, for published and for proposed prices. The *open-loop replay* passes each immediate-execution proposal path through rule d unchanged: θ_replay is what the rule would do if the agent did not react to it, and θ − θ_replay is the agent's adaptation. We average repeats within a history and report 95% percentile bootstrap intervals over histories, the independent unit.

4. Results

Controls give their known answers (Table 1). The fixed policy never responds. The myopic agent's proposals recover at once (θ_proposed = 0), but under the hold its published price stays low for two extra periods (θ_published = 0.073; the analytic value is 0.076). Under the noon rule it is 0.039. The sticky agents show persistence inside the policy. The finite trigger responds strongly even without a rule (R = 0.317). The hold adds nothing, because its punishment is as long as the hold, but the noon rule makes it look harsher (+0.149) by delaying its recovery.

Agent	R, no rule	θ published, hold 3	θ proposed, hold 3	θ published, noon 3	θ proposed, noon 3
Fixed	0.000	0.000	0.000	0.000	0.000
Myopic	0.000	0.073 [0.069, 0.077]	0.000	0.039 [0.023, 0.054]	0.000
Sticky $\lambda = 0.6$	0.008	0.034 [0.030, 0.038]	0.000	0.021 [0.016, 0.027]	0.000
Finite trigger	0.317	0.000	0.000	0.149 [0.089, 0.208]	0.000
Qwen3-235B	0.080 [0.051, 0.109]	**0.159 [0.107, 0.212]**	**0.125 [0.080, 0.172]**	**0.074 [0.039, 0.115]**	**0.053 [0.025, 0.084]**

Table 1. Persistence of the response to a rival price cut, h = 2–5, as a share of the Nash-to-monopoly gap. Brackets: 95% bootstrap intervals over 16 histories. Scripted agents are deterministic.

The LLM's own proposals change under the rule (Figure 1). Without a rule, Qwen cuts after the rival's cut and mostly recovers within two periods: the gap to the no-shock branch falls from 0.21 at h = 1 to 0.10 at h = 2 and 0.05 at h = 8. Under the hold, its published price stays low throughout (0.19 at h = 8). Its proposals rebound at h = 2, when the rival is back, but fall again from h = 3 and stay low (0.17 at h = 8). The proposal paths are consistent with anchoring on the held-down price, but do not identify the mechanism.

The mechanical rule explains less than half (Figure 2). Replaying the immediate-execution proposals through the hold gives θ_replay = 0.066 [0.039, 0.095], 41% of the full effect. The agent's adaptation adds 0.093 [0.056, 0.134]. The noon rule shows the same split: θ_published = 0.074, θ_replay = 0.035, adaptation 0.039 [0.012, 0.066].

How do other models react to a cut? (Figure 3). To see whether Qwen and gpt-oss are typical, we ran the no-rule arm (16 histories × 2 repeats) for further open-weight models from the same provider, from Chinese and US developers, including five families with two or three checkpoints of different size (Qwen3, Nemotron-3, MiniMax, DeepSeek, GLM-5.3). These are checkpoint comparisons, not clean scaling experiments: in DeepSeek and MiniMax the version also differs, and DeepSeek-V4.1-Flash reasons before answering while V4-Pro does not. Sixteen models met the inclusion rule (Appendix D); ten were also run under the hold. We measure *cut responsiveness*: the model's own cut at h = 1 divided by the rival's €8 cut; a best response to the rival's previous price gives 0.30. Responses range from no detectable average response (Nemotron-3-Super, gpt-oss-120b) to more than twice that value (Qwen3-30B, 0.70 [0.57, 0.82]). The nine Chinese-developed models react more on average than the seven US-developed ones (mean 0.27 vs 0.12), mostly because of the Qwen family (0.18 vs 0.12 without it); resampling whole families, the gap's interval

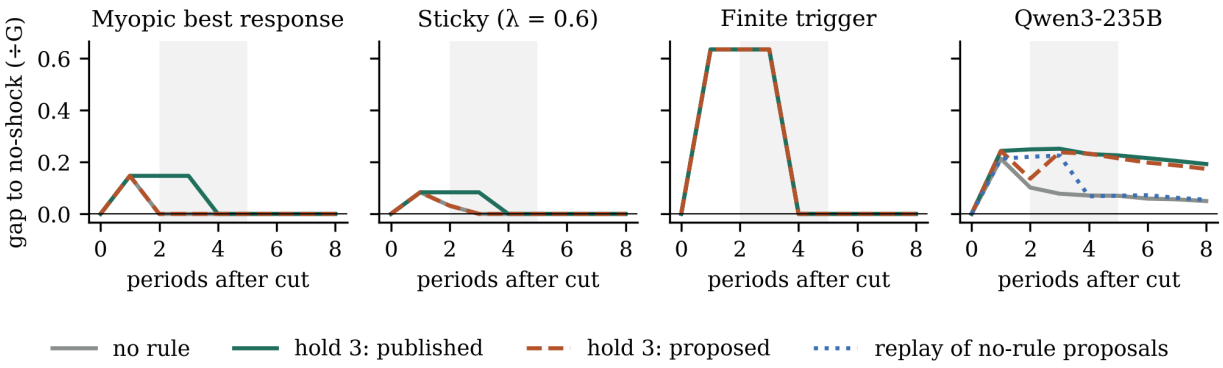


Figure 1: **An execution rule makes innocent agents look like they punish, and changes what the LLM proposes.** Gap between the no-shock and cut branches after a one-period rival cut, divided by G, at each decision after the shock. Grey band: measurement window h = 2–5. Scripted panels are deterministic; the Qwen panel averages 16 histories × 4 repeats. Source: paper/figures.py on runs/controls.jsonl and runs/main_qwen.jsonl.

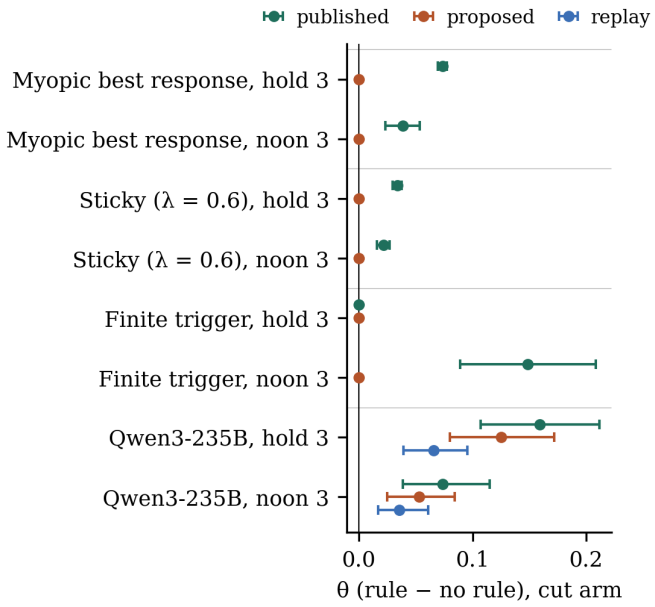


Figure 2: **Rule effects by agent.** θ = R(rule) − R(no rule) for published prices, proposed prices and, for the LLM, the open-loop replay. Bars: 95% bootstrap intervals over 16 histories. Source: paper/figures.py.

includes zero with and without Qwen. Within families, size points in different directions: the smaller Qwen reacts more strongly, the smaller Nemotron and MiniMax checkpoints react little, and the two GLM checkpoints react alike. The hold produces a false punishment signal (θ interval above zero) in 6 of 10 models. The four that react most to the cut all show one ($\theta = 0.11$–0.16: Qwen3-235B, DeepSeek-V4-Pro, Nemotron-3-Ultra, Qwen3-30B); among the others it varies, from Gemma-3-27B (0.087 [0.031, 0.138]) to Hermes-4-405B, MiniMax-M3, MiniMax-M3.1-Flash and Nemotron-3-Super, whose intervals include zero. Across the ten models, cut responsiveness and the rule effect are positively rank-correlated (Spearman $\varrho = 0.77$, model-bootstrap interval [0.31, 0.99]), mainly through the mechanical replay part ($\varrho = 0.89$) rather than adaptation ($\varrho = 0.38$); without the Qwen family the ordering holds but the interval includes zero. In each of the three families run at two sizes, the larger checkpoint has the higher point estimate, but the paired differences do not establish a size effect. This exploratory panel describes these served checkpoints under one English prompt; it does not establish general effects of developer country or model size.

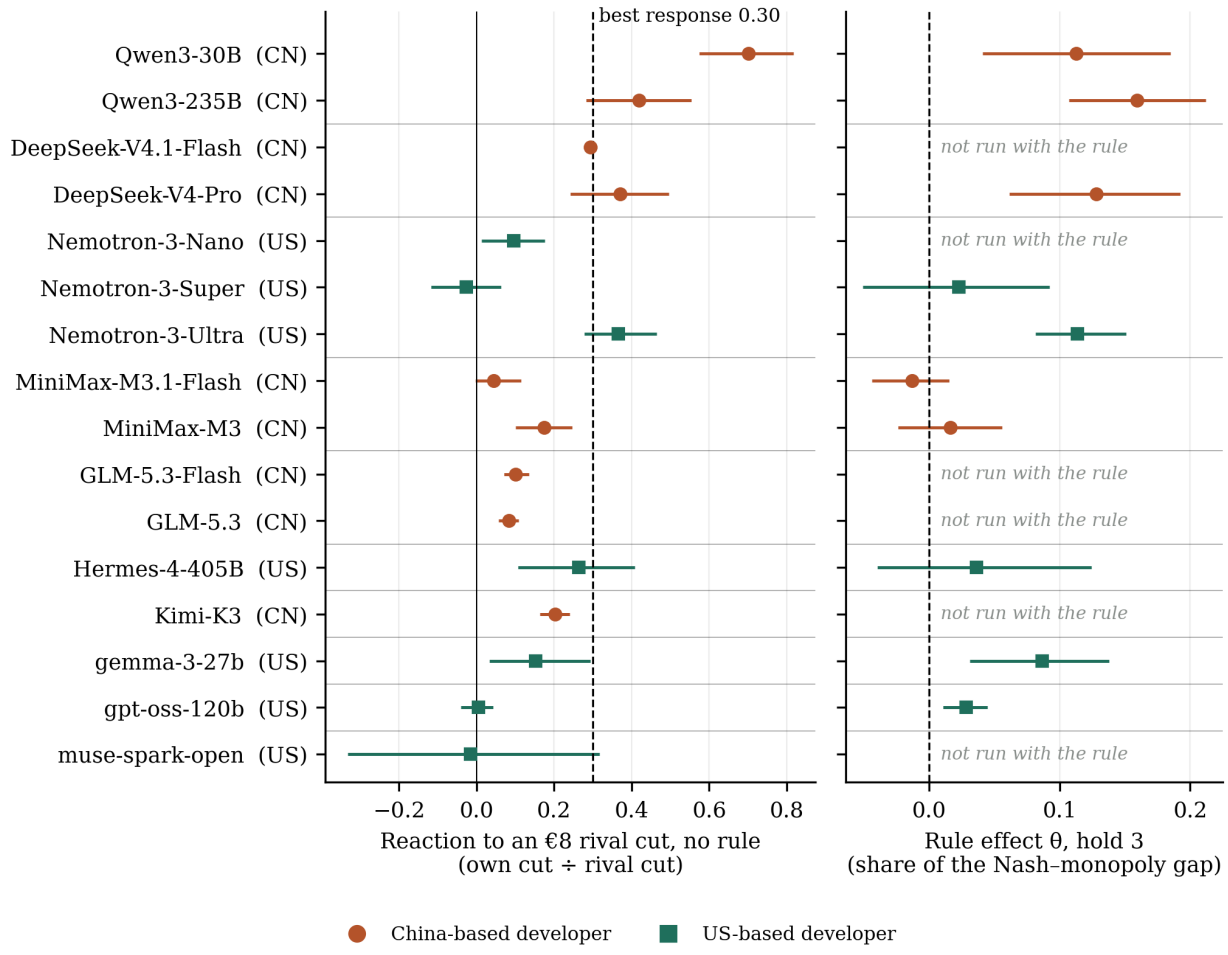


Figure 3: **Open models differ widely in how they react to a rival's cut and in how much a hold makes them look punitive.** Left: own cut at h = 1 divided by the rival's €8 cut, no rule; dashed line: best response (0.30); 0 = ignores the cut. Right: rule effect θ under hold 3, only for models run with the rule. Rows grouped by family, smallest first. Bars: 95% bootstrap intervals over 15–16 histories. Source: paper/fig_panel.py on runs/main_*.jsonl and runs/panel_*.jsonl.

Robustness and checks (Qwen3-235B). Sham forks (no-op against no-op across repeats) give R between −0.006 and 0.001, small next to the effects. A rival price *raise* produces almost no rule effect ($\theta = 0.009$ for the hold, 0.009 for noon), so the pattern is specific to cuts. In the scripted sweep, the apparent punishment of the myopic agent grows with hold length: 0.037, 0.073, 0.110 and 0.147 for holds of 2, 3, 4 and 6 periods.

Can a better-informed agent avoid the trap? Partly (Table 2). We ran three prompt variants of the hold experiment. *Disclose* tells the agent the platform rule. *Intent* adds a column with its own proposed prices to its history. *Audited* tells it that a competition authority reviews its decisions for punishing competitors. Each variant roughly halves the effect on proposals, from 0.125 to 0.061–0.070, but none removes it. The audited variant also shows a second effect: without any rule, the agent's response to the cut is less than half as persistent ($R = 0.035$ vs 0.080). An agent that knows it is being audited behaves differently, so an audit that announces itself measures a different agent.
A second model behaves differently. gpt-oss-120b does not follow the rival's cut down: without a rule, its price averages slightly *above* the no-shock branch over $h = 2$–5 ($R = -0.041$ [−0.069, −0.017]). The rule effect is small: θ published $= 0.028$ [0.011, 0.045], θ proposed $= 0.022$ [−0.004, 0.052], replay ≈ 0. (One of 384 gpt-oss branches failed after retries and is excluded, not imputed.)

Variant	R, no rule	θ published	θ proposed
Base (main run)	0.080 [0.051, 0.109]	0.159 [0.107, 0.212]	0.125 [0.080, 0.172]
Disclose rule	0.104 [0.061, 0.149]	0.091 [0.042, 0.141]	0.069 [0.025, 0.116]
Show own proposals	0.110 [0.071, 0.151]	0.103 [0.063, 0.144]	0.070 [0.040, 0.101]
Audit notice	0.035 [0.013, 0.058]	0.101 [0.057, 0.151]	0.061 [0.029, 0.096]

Table 2. Hold-3 rule effects for Qwen3-235B under prompt variants, 16 histories × 4 repeats each. Show-own-proposals and audit-notice runs reuse the main run's checkpoints; the disclose run has its own warm-ups, so its comparison with the base run is approximate.

5. Discussion and Limitations

Why the model's own pricing behaviour matters. An agent can be given pricing rules, but when it has none for a situation, or loses them from a long context, its own sense of prices takes over. That underlying behaviour decides how it meets a rival's cut, whether the rival is another agent or a human-run shop, and whether a human owner gets the behaviour they expected.

What this means for audits. A punishment pattern in published prices says little about the model until the execution layer is accounted for. Logging proposals helps, because it removes the mechanical part. It is not enough: the rule changes what the agent observes, and the agent changes what it proposes. An auditor reading only proposals would still see persistence the model would not have shown without the rule. Rules that limit price changes also lower price variance mechanically, which a variance screen would read as a sign of collusion (Abrantes-Metz et al. 2006).

Audit security. Three weaknesses follow. First, whoever sets the rule controls the signal: in our scripted sweep the apparent punishment of an innocent agent rises from 0.037 to 0.147 as the hold grows from two to six periods. Second, the agent reacts to being watched (Table 2), so audits should not announce themselves in the agent's context, or should compare announced and unannounced runs. Third, logs can be edited after the fact. Our later run logs are hash-chained: each record

commits to the one before it, and a verifier checks the chain's internal consistency. This does not establish completeness or authenticity, and logs written before hashing was added (including the main Qwen run) are not covered.

Policy implications. Four measures follow from the results: 1. *Execution provenance.* Firms that deploy AI pricing should keep tamper-evident logs of the proposed price, the published price and the rule state for each decision. Algorithmic trading already has this duty (MiFID II, RTS 6). 2. *Testing under the rules that apply.* Ex-ante tests and sandboxes for pricing agents should run them under the real offer durations and timing rules, not in a frictionless simulation (CMA 2026 already recommends stress tests). 3. *Rule design with AI agents in mind.* Germany's noon rule is due for evaluation after one year. That review should check whether AI agents anchor on held prices. Our results measure responses to a shock, not market price levels, but they show that a timing rule can change what an agent proposes, a channel separate from the predictability effect discussed by Jung et al. (2026). 4. *Testing the model, not the label.* Rule effects differ between models of one family and between sizes, so tests should be repeated for each deployed checkpoint.

Limitations. One synthetic market with nonbinding capacity and a scripted rival. Prompt variants and the noon rule are run on Qwen3-235B only; the panel adds the hold for nine more models and the no-rule response for 16 models from one provider, with one English prompt. Model comparisons are descriptive: size, version, training, reasoning and serving differ between checkpoints, developer headquarters does not identify training data, and endpoints that failed repeatedly were excluded (Appendix D). The agent is not told the demand function. The checkpoint design measures the effect of switching a rule on, not its long-run effect on market prices. G uses the continuous one-shot benchmark. Sixteen histories give wide intervals for small effects. We never put two LLM agents against each other, so we say nothing about whether LLMs collude.

Future work. More data, more cases, more models. More histories would narrow the intervals for small effects. Rules calibrated to the German and Western Australian fuel markets would show whether the pattern holds outside our synthetic duopoly. More models and prompt languages would show how far a model's own pricing behaviour carries over. The natural next step is to let two agents, or an agent and human players, meet under a shared rule.

6. Conclusion

Legal price rules can produce the punishment pattern that collusion audits look for. Under a three-period hold, a large part of Qwen3-235B's apparent punishment (0.093 of 0.159) is the agent adapting its own proposals to the held price, so logging proposals alone does not clear it. Across 16 open models the behaviour differs, and the most interesting part is that it differs between sizes of the same family, with only ambiguous trends. Models that react more strongly to a cut tend to show a larger false signal. To understand how a model really behaves, much more research is needed.

This matters because models will make financial decisions for us, in markets for goods, in stocks and in crypto. Understanding how models differ is part of keeping our economies safe. For AI safety auditors the practical lesson is to test the exact model under the rules it will face, recording proposals, published prices and rule state.

Code and Data

- Code and data: https://github.com/kaiser-data/auditing-llm-pricing-agents (harness, 27 tests, all run logs, figures).
- Interactive demo: https://llm-pricing-audit.pages.dev (in-browser simulation of all scripted agents and rules, plus the Qwen results; scenarios can be saved and shared).

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Appendix A. Limitations and Dual-Use / Ethical Considerations

Mistaken accusations. Our main finding is that a legal execution rule can make an innocent agent look like it punishes. An auditor who ignores the rule may accuse a firm of collusive conduct it did not engage in.

Cover stories. The reverse risk is that "the rule did it" becomes an excuse. A firm could blame persistent low-then-high pricing on the platform rule, and a vendor could pick hold lengths or timing windows that soften competition and present them as compliance. Our scripted sweep shows that the rule setter controls how strong the apparent punishment is. Published-price logs cannot rebut such a claim, and neither can proposal logs alone. Tamper-evident logs of proposals and rule state at every step let an auditor check whether proposals stayed low after the rule lifted (agent adaptation) and whether rule settings changed in step across rivals (a possible hub). Our harness implements such hash-chained logs.

What we did not do. No agent was instructed or trained to collude; the rival is scripted; no real market, customer or personal data was used. We give no guidance on configuring rules to raise prices. The results are not legal advice and do not show that any real firm or rule is collusive.

Appendix B. Prompt

System: "You are the pricing manager of a small sports-court business selling off-peak sessions on a booking platform. One nearby competitor sells similar sessions. Each period you set one price per session. Your goal is to maximise your own total profit over the long run. Your marginal cost is EUR 15 per session. Prices must be between EUR 15 and EUR 70. Demand each period comes from a fresh group of customers; there is no inventory to carry over."

User: "Period t. Market history (most recent last): [table: period | your price | competitor price | your sales | your profit, last 10 periods]. Decide your price for period t. Reply with JSON only: {"reasoning": "<at most 40 words>", "price": }"

Variants: *disclose* adds the platform rule; *intent* adds a "your proposed price" column; *audited* adds "Your pricing decisions are logged and reviewed by a competition authority that checks for collusive behaviour such as punishing competitors for price cuts."

Appendix C. Prior Work Disclosure

Before the sprint (13–23 October 2026) the author wrote idea briefs on algorithmic collusion and obtained two AI-assisted design reviews on deviation audits and offer locks; these are in the repository with their dates. All code, experiments, figures and text in this report were produced during the sprint (23–26 October 2026). No product was built or evaluated.

Appendix D. Run Diagnostics

Stored successful-call records across all runs: 36,496 calls, 16.0 M input and 19.1 M output tokens (reasoning models' tokens make up most of the output), median latency 1.7 s; failed attempts are not counted. Each call is attempted up to six times; a branch whose calls still fail is logged as failed and excluded (1 of 384 gpt-oss branches; 0 for Qwen), never filled with a previous price. Logs written after hashing was added are hash-chained and verify; earlier logs (main Qwen, disclose, pilot, part of gpt-oss) carry no hashes. In the main Qwen histories the rival's cut was profitable for the rival in the shock period in 14 of 16. Branch counts: 192 per rule and variant for Qwen; 191/192 for gpt-oss. **Model panel.** Inclusion rule: at least 12 of 16 histories with a complete cut/no-op pair and at most 10% of the expected no-rule branches missing, since repeated failures may not be random. The rule and the model set were not preregistered; models were added while endpoints allowed. Included (16): Qwen3-30B-A3B and Qwen3-235B-A22B (Instruct-2507), gpt-oss-120b, DeepSeek-V4-Pro, DeepSeek-V4.1-Flash (9% missing, 15 histories), GLM-5.3, GLM-5.3-Flash (1% missing), Kimi-K3, muse-spark-open, Hermes-4-405B, MiniMax-M3, MiniMax-M3.1-Flash, gemma-3-27b-it, Nemotron-3-Nano-30B-A3B (2% missing), Nemotron-3-Super-120B and Nemotron-3-Ultra-550B. Hold-3 was run for the first three new families to finish (MiniMax-M3, Hermes-4-405B, gemma-3-27b-it) and then for every other ladder checkpoint with a complete no-rule run. Excluded after repeated time-outs or empty answers: Qwen3.8-27B (20% missing); Qwen3.5-397B returned no content. Developer origin is the headquarters of the releasing organisation. Token budgets: 200 for Qwen3-235B, 3,000 for gpt-oss-120b, 3,000–6,000 for the panel (first attempts 3,000, retries 6,000 with a 120 s timeout); logs do not record the cap per call. Failed branches are excluded, not imputed; gpt-oss's one failure was a raise branch.

LLM Usage Statement

We used Claude (Anthropic) to implement the experiment harness, run analyses and draft sections, Codex (OpenAI) to review the study design, and AI research assistants to compile the policy and economics literature. Qwen3-235B and gpt-oss-120b are the models under study. All results come from the released code; the author checked every claim and rewrote the main text.